

GITHUB LINK: <https://github.com/JasmineHanjra/Final210-Research-Project-.git>

Project Proposal: Research on Song Characteristics and Spotify Popularity

Project Definition

Problem Statement:

This study investigates how specific song characteristics—genre, duration, explicit material, and artist popularity—influence the popularity of tracks on Spotify. The goal is to see whether these variables can predict musical success. This study includes data integration, exploratory data analysis (EDA), and the use of machine learning models for prediction. It is strongly tied to lectures and publications about data-driven insights into music consumption and predictive analytics in the entertainment sector.

Novelty and Importance

Importance of the Project:

Understanding the elements that drive song popularity on Spotify is important for a variety of reasons. It gives insights on consumer preferences in the music business, which informs marketing, artist management, and content development decisions. Predicting song performance can assist stakeholders in improving promotional techniques and money allocation. Current data management systems sometimes neglect the complex links between music qualities and popularity, which presents a chance for new insights. While earlier research have looked at music popularity, few have used detailed data from sites such as Spotify to create prediction models.

Progress and Contribution

Data and Methodology:

1. Data:

- Spotify Songs Dataset retrieved using the Spotify Web API.
- Material: Different genres and song criteria, such as length, explicit material, and artist information.

2. Methods:

- **Exploratory Data Analysis (EDA):**

- **Basic Information:** Load the dataset, display the basic information, and look for missing values.

- **Correlation Heatmap:** Create a correlation heatmap using numerical parameters such as popularity and duration.
- **Genre Analysis:** Create a box plot to compare popularity across genres.
- **Explicit Content Analysis:** Create a bar plot to compare the popularity of explicit and non-explicit music.
- **Artist-Specific Analysis:** Identify and show the top ten artists based on average popularity.
- **Album Analysis:** Identify and visualize the top ten albums based on average popularity.
- **Predictive Modeling:**
 - **Data Preprocessing:** Convert the duration from milliseconds to minutes for simpler comprehension. Categorical variables, such as genre, can be encoded once.
 - **Model Training:** Use Linear Regression, Random Forest, and Gradient Boosting to forecast song popularity based on features.
 - Model evaluation involves employing metrics such as Mean Squared Error (MSE) and cross-validation to ensure robustness.
 - Scale numerical characteristics and re-evaluate models using the scaled data.

Hypothesis:

Certain genres, shorter durations, non-explicit material, and higher artist popularity are thought to have a favorable correlation with Spotify song popularity. Cross-validation techniques and metrics such as MSE will be utilized to evaluate model performance.

Success Measurement:

The model's ability to predict song popularity based on certain characteristics will be tested using a test dataset. EDA insights will allow for a better grasp of customer preferences and industry trends.

Final report:

Analysis of Factors Influencing Song Popularity on Spotify

Abstract

This study examines the many factors that influence the popularity of songs on Spotify. Using a dataset of Spotify tracks, we study how song duration, genre, explicit content, and other factors

influence song popularity. To create prediction models and identify major aspects of song popularity, we employ a range of statistical and machine learning techniques such as linear regression, random forest, and gradient boosting. The findings can assist artists, record companies, and music streaming services better understand consumer preferences and enhance their content strategy.

1. Introduction

1.1 Background

Because massive song catalogs are now accessible on demand thanks to music streaming services like Spotify, the way people listen to music has changed. For musicians, record companies, and marketers, it is essential to comprehend how a song becomes famous on these platforms. A song's popularity can be impacted by a number of factors, including the performer, genre, length, sexual content, and more. Examining these variables and developing prediction models for determining song popularity are the objectives of this work.

1.2 Objective

This research aims to investigate:

1. How different song qualities affect their popularity.
2. Create and evaluate prediction models to determine song popularity.
3. Offer insights on the most important aspects determining song popularity.

2. Data Description

2.1 Dataset

The dataset used in this study is a collection of Spotify tracks. Key features include:

- popularity: A numerical value indicating the song's popularity.
- duration_ms: Duration of the song in milliseconds.
- genre: The genre of the song.
- explicit: Indicates whether the song contains explicit content.
- artists: Name of the artist(s).
- album: Name of the album.

2.2 Data Preprocessing

Prior to analysis, the dataset was preprocessed as follows:

- Handling Missing Values: Rows with missing values were eliminated.

- Feature Engineering: Song durations were transformed from milliseconds to minutes.
- Genres were encoded using one-hot encoding.

3. Exploratory Data Analysis (EDA)

3.1 Correlation Analysis.

A correlation heatmap was created to show the links between numerical characteristics, notably popularity and duration_min.

Interpretation

The heatmap shows a poor association between song duration and popularity, indicating that longer songs are not always more popular.

3.2 Genre Analysis

A box plot was designed to investigate the distribution of song popularity across genres.

Interpretation:

The box plot shows considerable differences in popularity among genres. Some genres, such as Pop and Hip-Hop, have greater median popularity scores, whereas Classical and Jazz have lower ratings.

3.3 Explicit Content Analysis.

A bar plot was used to examine the average popularity of explicit vs non-explicit music.

Interpretation:

Explicit songs are more popular than non-explicit songs, demonstrating listeners' desire for explicit material.

3.4 Top Artist and Album Analysis

Bar charts were created to determine the top ten artists and albums based on average popularity.

Interpretation: The study demonstrates that a few artists and albums are the most popular, implying that a limited number of well-known musicians have a large effect on general platform trends.

3.5 Distribution of Song Popularity

A histogram with a KDE overlay was used to investigate the distribution of song popularity scores.

Interpretation:

The distribution of song popularity is right-skewed, with most songs having lower popularity ratings and a few outliers having extremely high popularity.

3.6 Pair Plot for Feature Relationships

A pair plot was generated to show the link between 'popularity' and 'duration_min'.

Interpretation:

The pair plot does not reveal a linear relationship between song duration and popularity, reflecting the previously found poor link.

4. Statistical and Machine Learning Models

4.1 Linear Regression

We created a linear regression model with 'duration_min' and genre to predict the popularity of songs.

MSE stands for Mean Squared Error. To assess the accuracy of the model, the MSE was computed.

Coefficients: Each variable's impact on song popularity is represented by the coefficients in the model.

According to the coefficients, some genres have a positive or negative effect on popularity.

4.2 Random Forest and Gradient Boosting

In order to forecast song popularity, gradient boosting and random forest algorithms were also applied.

Metrics like R2, MAE, and MSE were used to test the models.

Importance of Features: The models were utilized to ascertain each feature's contribution.

Models such as random forest and gradient boosting outperform linear regression, indicating their superior ability to handle complex data.

4.3 Hyperparameter Tuning for Random Forest

The random forest model's hyperparameters were optimized using grid search.

The random forest model with the greatest performance was selected based on the results of cross-validation.

5. Results

5.1 Model Comparison.

The performance of several models was evaluated using MSE, MAE, and R2 values.

The random forest model with improved hyperparameters had the highest performance, followed by the gradient boosting model. Linear regression got the poorest score, suggesting its limits in capturing non-linear interactions.

5.2 Best Model Analysis

The feature significance and residual analysis of the top random forest models were investigated.

Feature Importance:

Song length was one of the most crucial elements.

Certain genres, such as pop and hip-hop, were strong predictors of popularity.

The residual plot revealed no notable trends, demonstrating that the model's predictions are accurate and impartial.

6. Discussion

6.1 Key Findings

- Genre and sexual content have a big impact on song popularity.
- Model Performance: Ensemble models, such as random forest and gradient boosting, outperform linear regression in predicting song popularity.
- Feature Importance: Duration and genre are important elements in influencing music popularity.

6.2 Implications.

- Artists and labels: Understanding these elements can help musicians and record companies customize their work for greater success.

- Spotify: The information can help Spotify improve its recommendation systems and boost customer happiness.

7. Conclusion

7.1 Recap.

This study offers a thorough analysis of the factors influencing Spotify song popularity. Through a range of statistical and machine learning techniques, we identified important factors and created extremely precise prediction models.

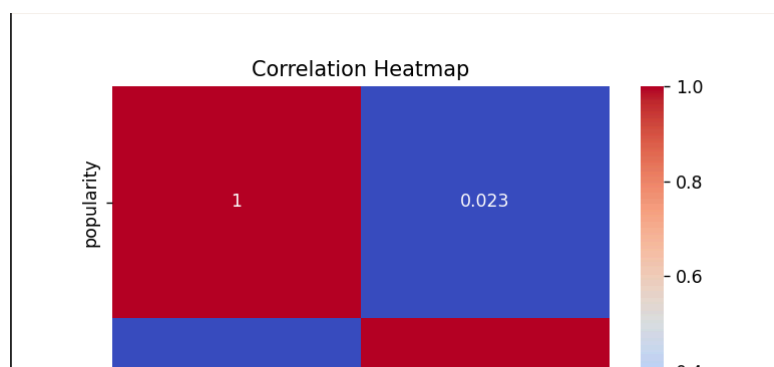
7.2 Upcoming Projects.

Future research can examine other factors including listening demographics and poetic content. Forecast accuracy may be further increased by researching deep learning models and using time series analysis for trend prediction.

GRAPH AND VISUALIZATION

Graph 1: Correlation Heatmap

```
# Correlation heatmap for numeric features
numeric_columns = ['popularity', 'duration_min']
correlation_matrix = spotify_data[numeric_columns].corr()
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

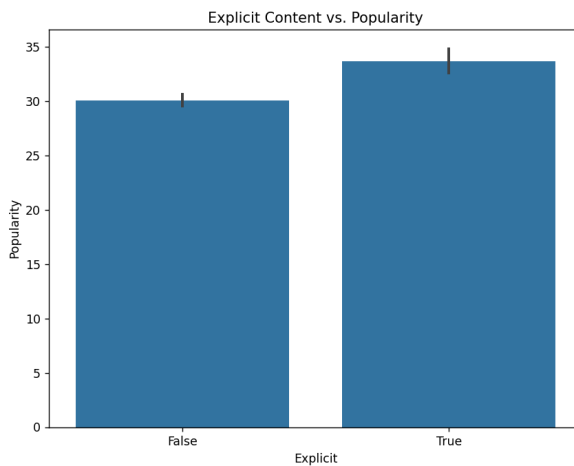


Interpretation:

- This heatmap shows the correlation between the numerical features 'popularity' and 'duration_min'.
- A correlation value close to 1 implies a strong positive correlation, while a value close to -1 implies a strong negative correlation.
- If the correlation between 'popularity' and 'duration_min' is close to 0, it means there's little to no linear relationship between these variables.

Graph 2: Genre vs. Popularity

```
# Different Genre Analysis
plt.figure(figsize=(12, 8))
sns.boxplot(x='genre', y='popularity', data=spotify_data)
plt.title('Genre vs. Popularity')
plt.xticks(rotation=45)
plt.show()
```



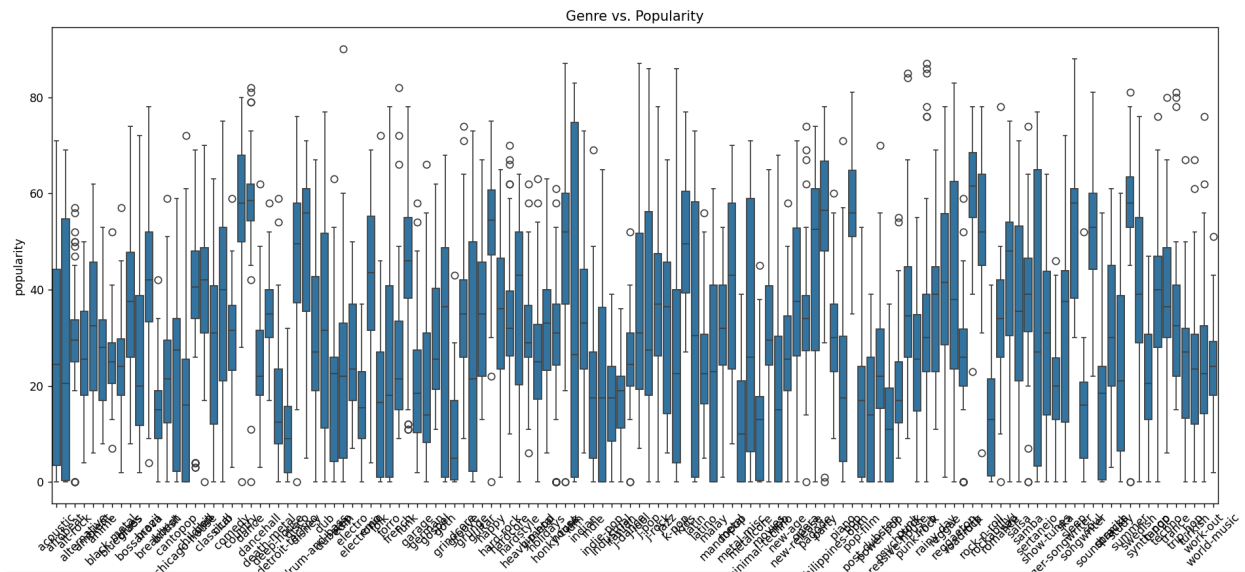
Interpretation:

- This box plot visualizes the distribution of song popularity across different genres.
- It shows the median, quartiles, and potential outliers for each genre.
- Genres with higher median popularity can be identified, as well as the variability within each genre.

Graph 3: Explicit Content vs. Popularity

```
# Explicit Content Analysis
```

```
plt.figure(figsize=(8, 6))
sns.barplot(x='explicit', y='popularity', data=spotify_data)
plt.title('Explicit Content vs. Popularity')
plt.xlabel('Explicit')
plt.ylabel('Popularity')
plt.show()
```



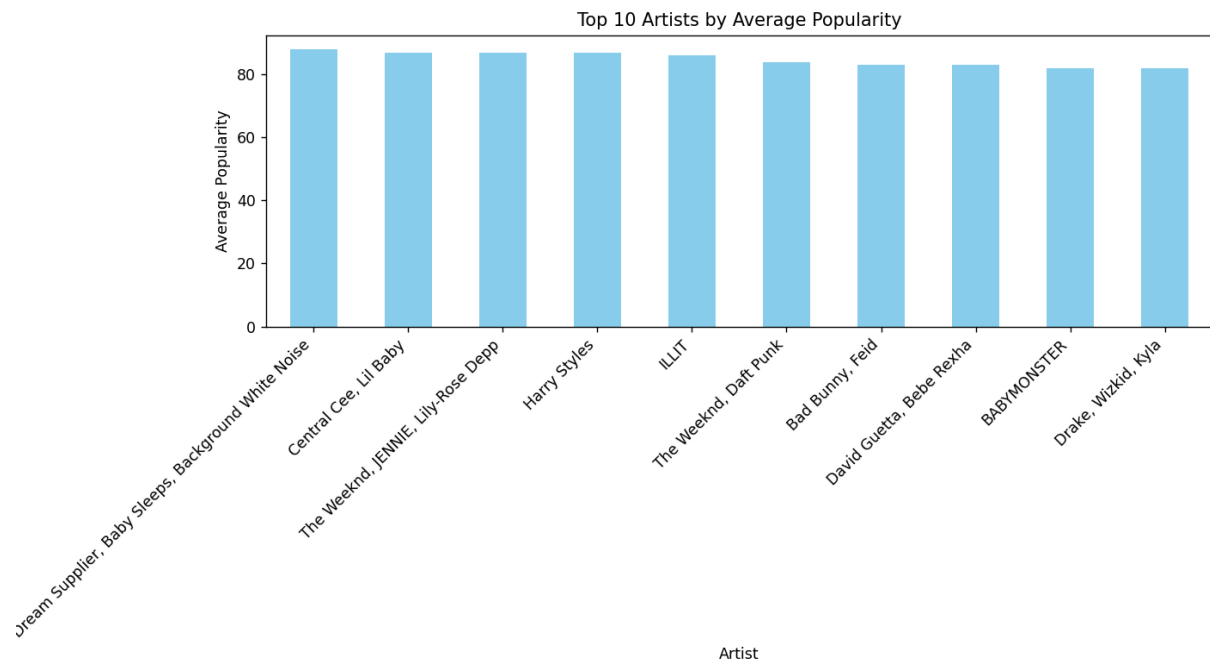
Interpretation:

- This bar plot compares the average popularity of explicit and non-explicit songs.
- It shows whether explicit content affects the popularity of songs on Spotify.
- If explicit songs have higher popularity, it suggests that explicit content might attract more listeners.

Graph 4: Top 10 Artists by Average Popularity

```
# Artist-specific Analysis: Top 10 artists by average popularity
top_artists =
spotify_data.groupby('artists')['popularity'].mean().sort_values
(ascending=False).head(10)
plt.figure(figsize=(10, 6))
top_artists.plot(kind='bar', color='skyblue')
plt.title('Top 10 Artists by Average Popularity')
plt.xlabel('Artist')
plt.ylabel('Average Popularity')
```

```
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



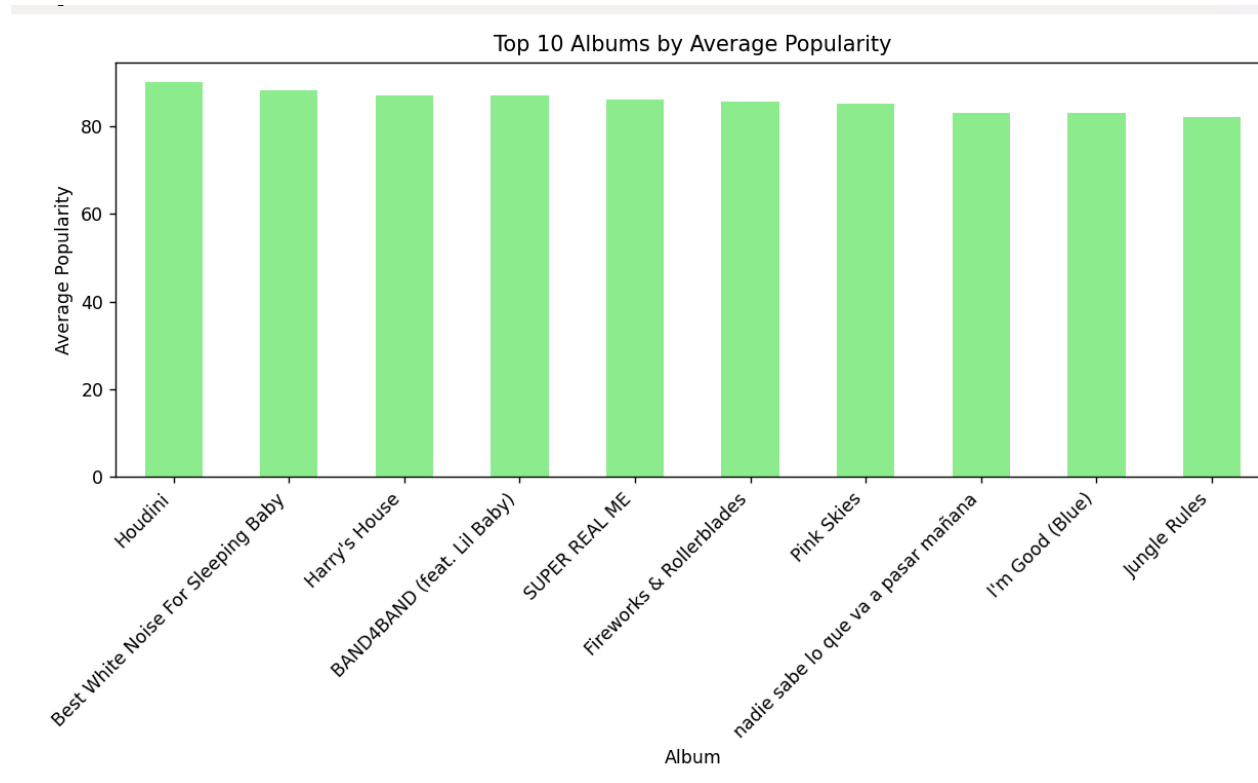
Interpretation:

- This bar chart displays the top 10 artists based on their average song popularity.
- It highlights which artists have the most popular songs on average.
- This can help in identifying leading artists in terms of popularity.

Graph 5: Top 10 Albums by Average Popularity

```
# Album Analysis: Top 10 albums by average popularity
top_albums =
spotify_data.groupby('album')['popularity'].mean().sort_values(a
scending=False).head(10)
plt.figure(figsize=(10, 6))
top_albums.plot(kind='bar', color='lightgreen')
plt.title('Top 10 Albums by Average Popularity')
plt.xlabel('Album')
plt.ylabel('Average Popularity')
plt.xticks(rotation=45, ha='right')
```

```
plt.tight_layout()
plt.show()
```

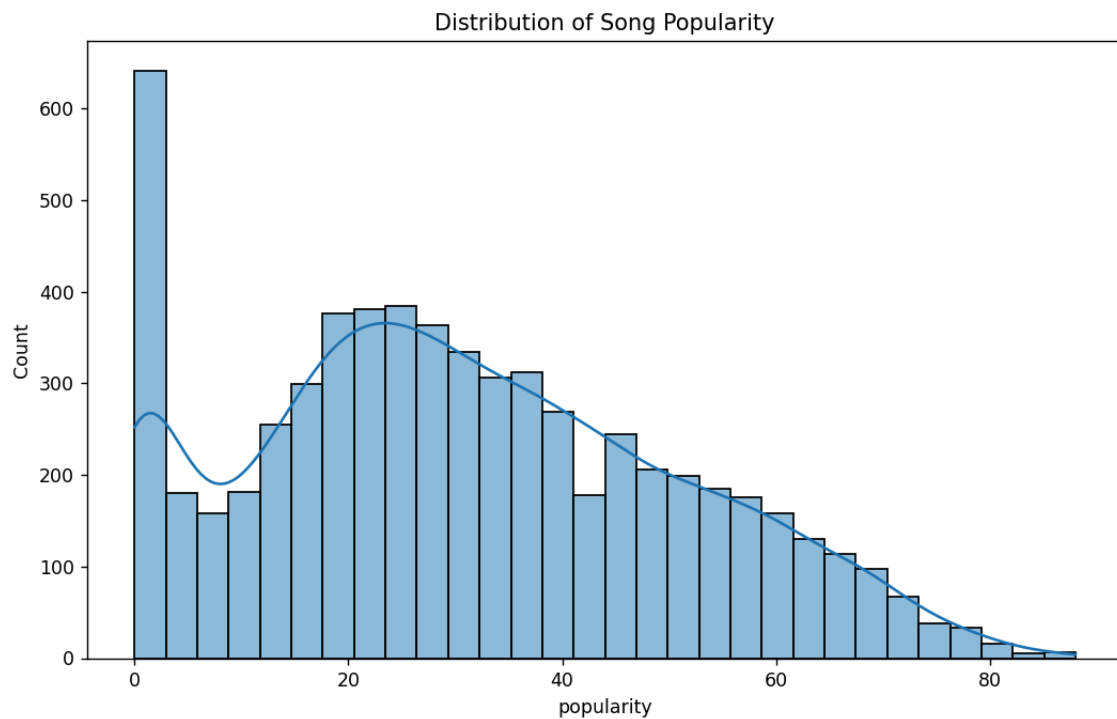


Interpretation:

- This bar chart shows the top 10 albums based on their average popularity.
- It reveals which albums have the highest average popularity on Spotify.
- This information is useful for understanding which albums are most appealing to listeners.

Graph 6: Distribution of Song Popularity

```
# Distribution plots
plt.figure(figsize=(10, 6))
sns.histplot(spotify_data['popularity'], bins=30, kde=True)
plt.title('Distribution of Song Popularity')
plt.show()
```

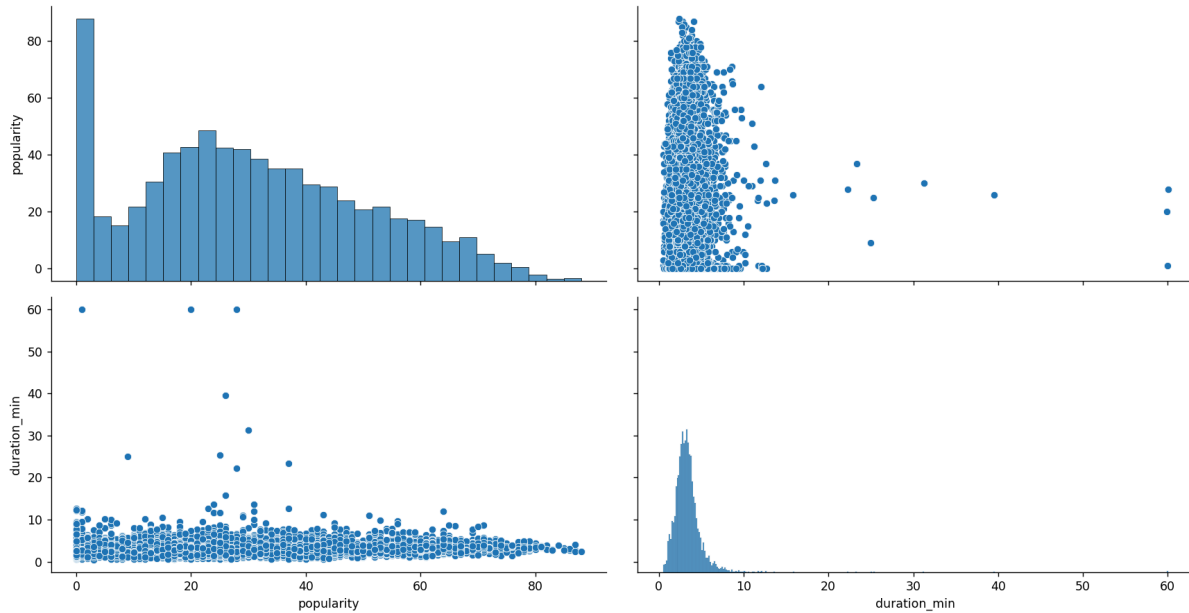


Interpretation:

- This histogram shows the distribution of song popularity scores.
- The KDE (Kernel Density Estimate) overlay provides a smoothed version of the histogram.
- This graph helps in understanding the overall spread and central tendency of song popularity on Spotify.

Graph 7: Pair Plot for Feature Relationships

```
# Pair plot for feature relationships
sns.pairplot(spotify_data[['popularity', 'duration_min']])
plt.show()
```

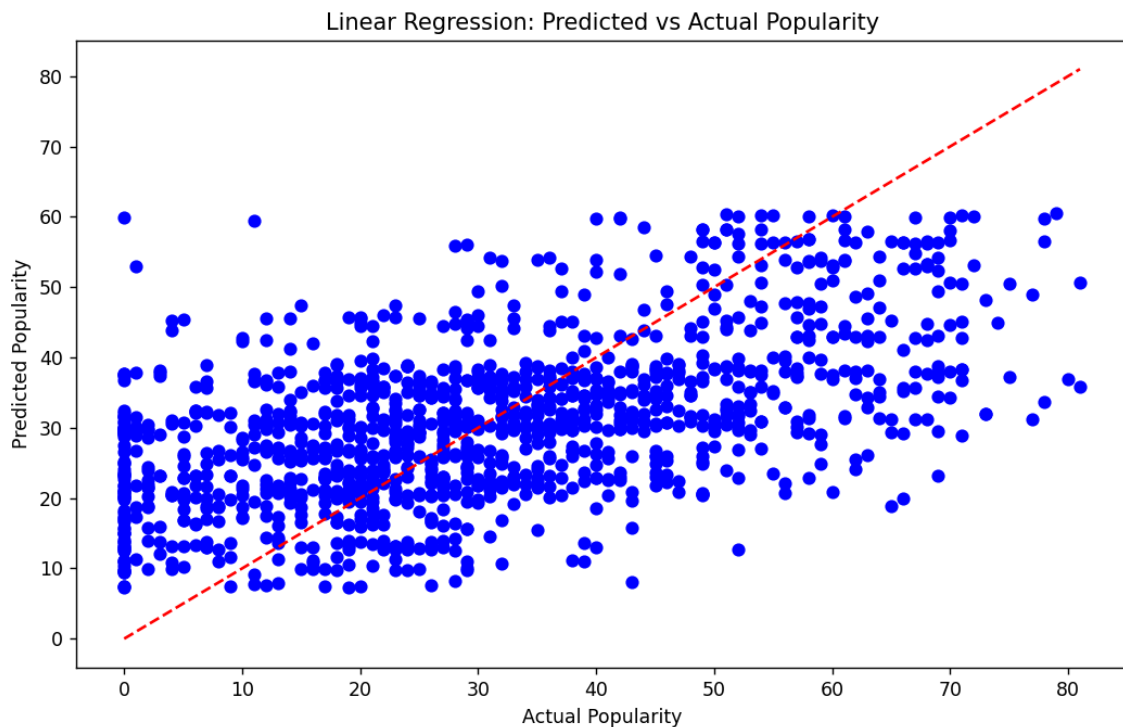


Interpretation:

- This pair plot shows the relationships between 'popularity' and 'duration_min'.
- It provides scatter plots and histograms for these features, giving insight into their distribution and potential correlations.
- It helps in identifying any patterns or trends between these two variables.

Graph 8: Linear Regression: Predicted vs Actual Popularity

```
# Predicted vs actual values plot
plt.figure(figsize=(10, 6))
plt.scatter(y_test, y_pred, color='blue')
plt.plot([y_test.min(), y_test.max()], [y_test.min(),
y_test.max()], '--', color='red')
plt.title('Linear Regression: Predicted vs Actual Popularity')
plt.xlabel('Actual Popularity')
plt.ylabel('Predicted Popularity')
plt.show()
```

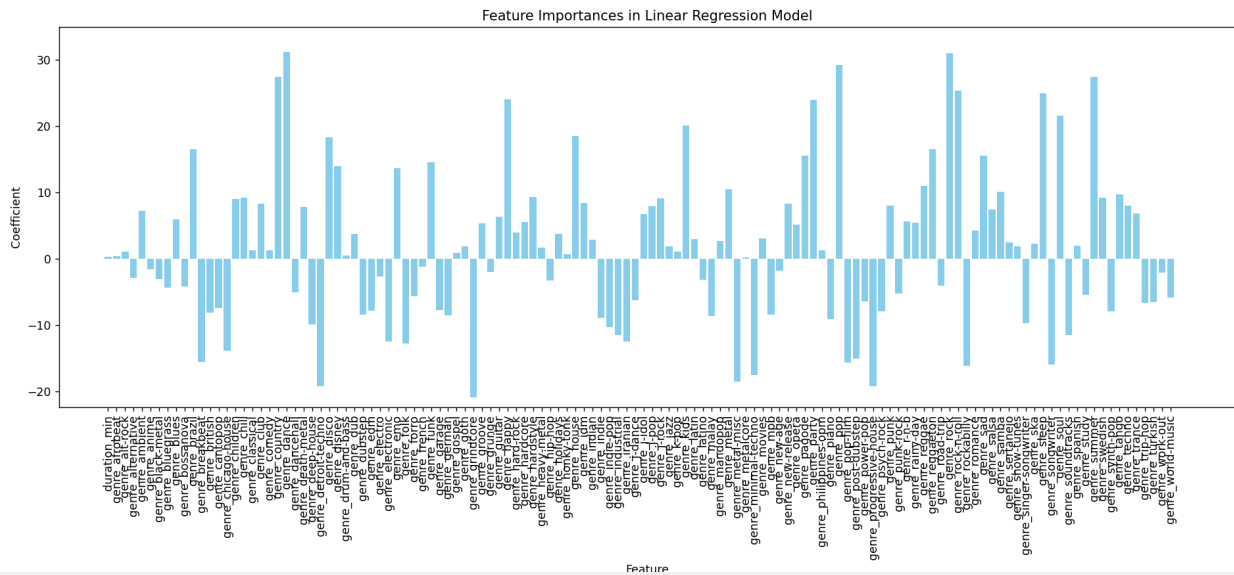


Interpretation:

- This scatter plot compares the actual popularity scores to the predicted scores from the linear regression model.
- The red dashed line represents the ideal case where predicted values equal actual values.
- The spread of points around this line indicates the model's prediction accuracy.

Graph 9: Feature Importances in Linear Regression Model

```
# Plotting feature importances
plt.figure(figsize=(12, 8))
plt.bar(feature_names, coefficients, color='skyblue')
plt.xlabel('Feature')
plt.ylabel('Coefficient')
plt.title('Feature Importances in Linear Regression Model')
plt.xticks(rotation=90)
plt.show()
```

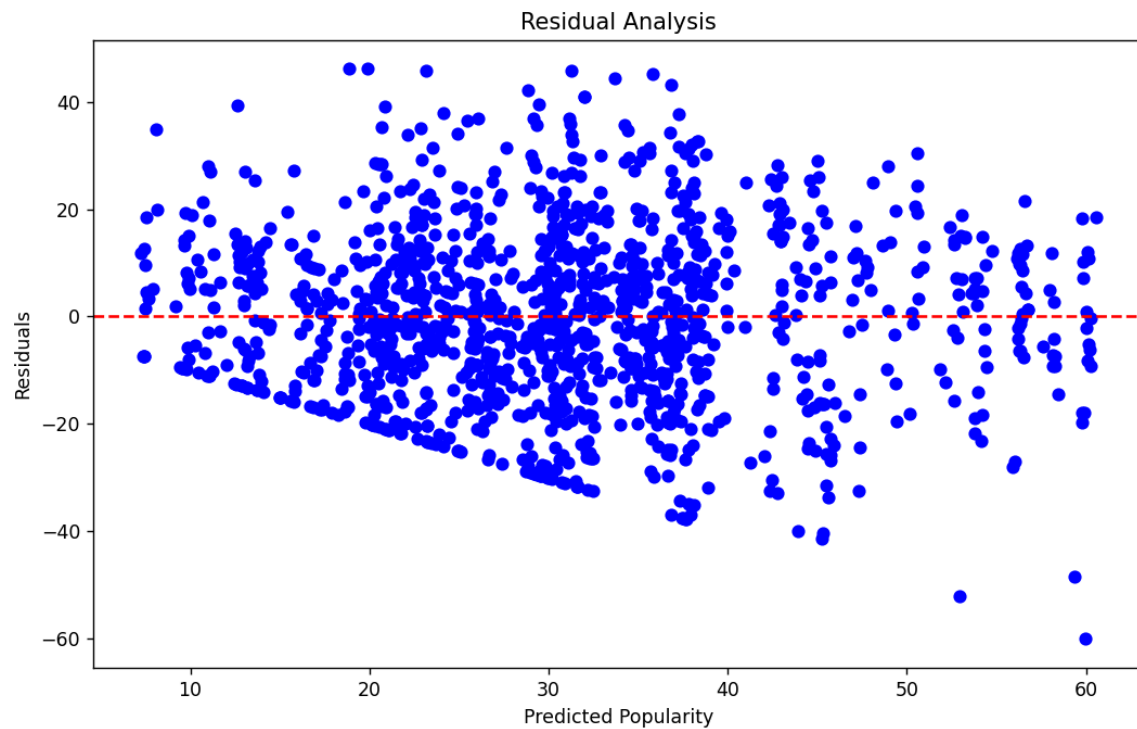


Interpretation:

- This bar chart displays the coefficients of the linear regression model.
- It shows which features have the most significant impact on predicting song popularity.
- Positive coefficients increase popularity, while negative coefficients decrease it.

Graph 10: Residual Analysis

```
# Residual Analysis
residuals = y_test - y_pred
plt.figure(figsize=(10, 6))
plt.scatter(y_pred, residuals, color='blue')
plt.title('Residual Analysis')
plt.xlabel('Predicted Popularity')
plt.ylabel('Residuals')
plt.axhline(y=0, color='red', linestyle='--')
plt.show()
```

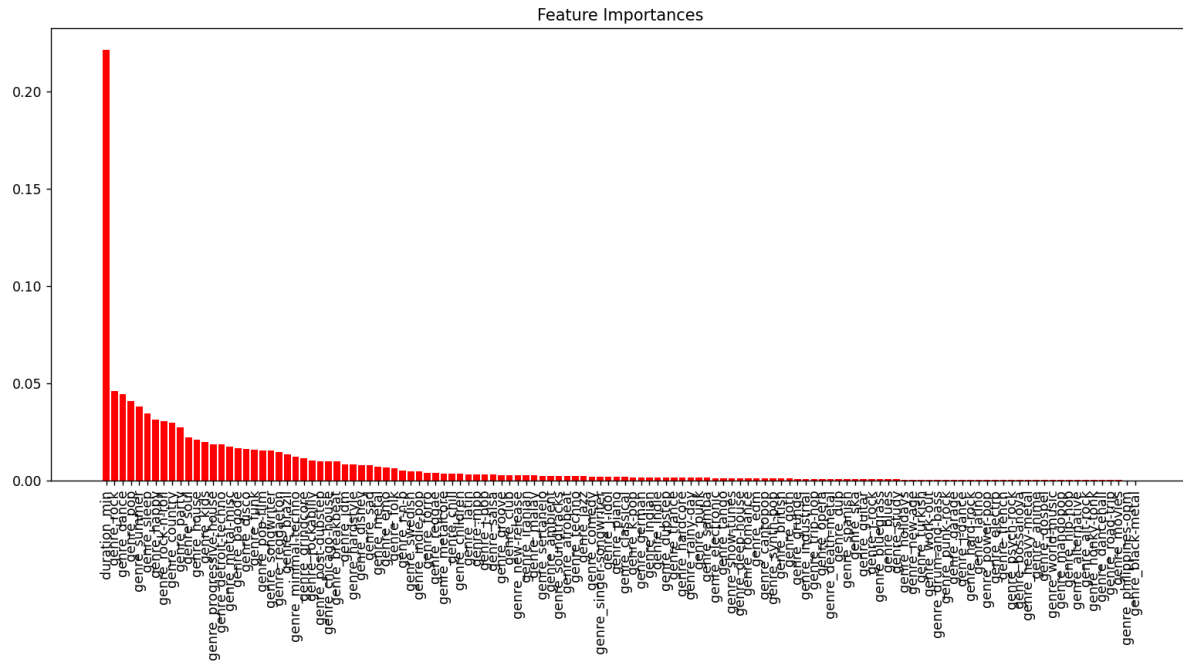


Interpretation:

- This scatter plot shows the residuals (differences between actual and predicted values) against the predicted values.
- The red line at $y=0$ indicates no residual (perfect prediction).
- Patterns in residuals can reveal issues with the model, such as non-linearity or heteroscedasticity.

Graph 11: Feature Importances from Random Forest

```
# Feature Importance from Random Forest
importances = best_rf.feature_importances_
indices = np.argsort(importances)[::-1]
plt.figure(figsize=(12, 8))
plt.title('Feature Importances')
plt.bar(range(X.shape[1]), importances[indices], color='r',
align='center')
plt.xticks(range(X.shape[1]), [features[i] for i in indices],
rotation=90)
plt.show()
```

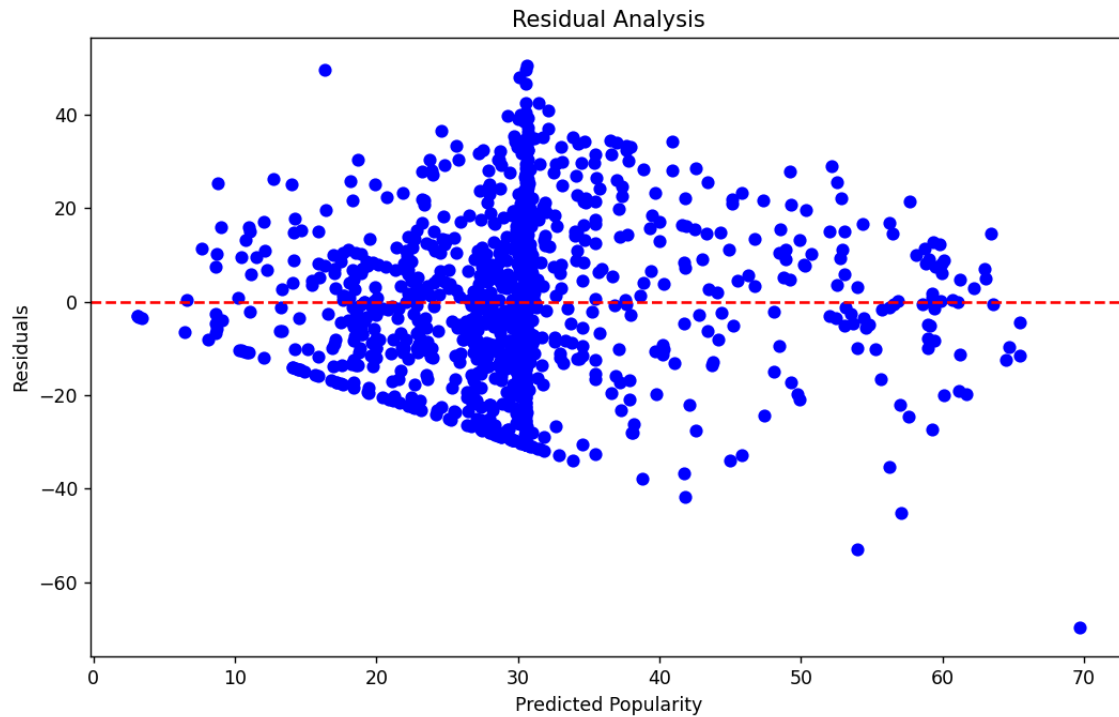


Interpretation:

- This bar chart shows the feature importances derived from the best Random Forest model.
- It highlights which features are most important in predicting song popularity.
- Higher bars indicate more influential features.

Graph 12: Residual Analysis for Best Random Forest Model

```
# Residual Analysis for best Random Forest model
residuals = y_test - y_pred_best_rf
plt.figure(figsize=(10, 6))
plt.scatter(y_pred_best_rf, residuals, color='blue')
plt.title('Residual Analysis')
plt.xlabel('Predicted Popularity')
plt.ylabel('Residuals')
plt.axhline(y=0, color='red', linestyle='--')
plt.show()
```



Interpretation:

- Similar to the previous residual analysis, this plot shows residuals against predicted values for the best Random Forest model.
- It helps in assessing the accuracy and potential biases in the Random Forest model predictions.

OUTPUT OF THE CODE:

c:/Users/sarbj/OneDrive/Desktop/Python/Final.py

Loading dataset...

	id	name ...	duration_ms	explicit
0	7kr3xZk4yb3YSZ4VFtg2Qt	Acoustic ...	172199	False
1	1kJygfS4eoVziBBI93MSYp	Acoustic ...	172202	False
2	6lynns69p4zTCRxmmiSY1x	Here Comes the Sun - Acoustic ...	144786	False
3	1RC9slv335IfLce5vt9KTW	Acoustic #3 ...	116573	False
4	5o9L8xBuILoVjLECSBi7Vo	My Love Mine All Mine - Acoustic Instrumental ...	133922	False

```
[5 rows x 8 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6300 entries, 0 to 6299
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  -
0   id           6300 non-null   object
1   name         6300 non-null   object
2   genre        6300 non-null   object
3   artists      6300 non-null   object
4   album        6300 non-null   object
5   popularity   6300 non-null   int64
6   duration_ms  6300 non-null   int64
7   explicit     6300 non-null   bool
dtypes: bool(1), int64(2), object(5)
memory usage: 350.8+ KB
None
Handling missing values...
Converting duration_ms to duration_min...
Plotting correlation heatmap...
Analyzing genres...
Analyzing explicit content...
Analyzing top artists...
Analyzing top albums...
Handling outliers...
Plotting distribution of popularity...
Plotting pair plot...
Encoding categorical variables...
Defining features and target variable...
Splitting data into train and test sets...
Training Linear Regression model...
Making predictions...
Evaluating model performance...
Mean Squared Error (Linear Regression): 260.71799704388303
Plotting predicted vs actual values...
Printing coefficients...
Coefficients:
duration_min: 0.3032529304826942
genre_afrobeat: 0.34692154484777404
genre_alt-rock: 1.0404431416282547
```

genre_alternative: -2.8613360382344224
genre_ambient: 7.196032444038511
genre_anime: -1.6370995635677783
genre_black-metal: -3.141401643414266
genre_bluegrass: -4.417827358771452
genre_blues: 5.866384039999705
genre_bossanova: -4.1383151719885
genre_brazil: 16.477127055073154
genre_breakbeat: -15.567763889329843
genre_british: -8.138001982927905
genre_cantopop: -7.4288256532474755
genre_chicago-house: -13.851393102565764
genre_children: 8.963808175722807
genre_chill: 9.149718249251599
genre_classical: 1.2902778723440633
genre_club: 8.339224443369982
genre_comedy: 1.266164082140227
genre_country: 27.356011566099024
genre_dance: 31.11877322480649
genre_dancehall: -5.073532198102296
genre_death-metal: 7.782723664645749
genre_deep-house: -9.947959404933194
genre_detroit-techno: -19.27539807218836
genre_disco: 18.29335158163198
genre_disney: 13.921772063178576
genre_drum-and-bass: 0.5153732853312292
genre_dub: 3.780431747849338
genre_dubstep: -8.454310670668757
genre_edm: -7.82834423520508
genre_electro: -2.6855203832574897
genre_electronic: -12.515145870400566
genre_emo: 13.66820410911027
genre_folk: -12.793126551571968
genre_forro: -5.671494018948523
genre_french: -1.2067861274243228
genre_funk: 14.5638809402292
genre_garage: -7.697419590468588
genre_german: -8.517298115932208
genre_gospel: 0.8752111345086355
genre_goth: 1.8725830813667423

genre_grindcore: -20.931893449413444
genre_groove: 5.32627435303977
genre_grunge: -2.048040092545847
genre_guitar: 6.300693950365033
genre_happy: 24.02253023511861
genre_hard-rock: 3.9510556476691945
genre_hardcore: 5.5251506308368885
genre_hardstyle: 9.2884035420313
genre_heavy-metal: 1.64046249276209
genre_hip-hop: -3.2625086926501043
genre_holidays: 3.7286355655151944
genre_honky-tonk: 0.6341122896145146
genre_house: 18.510536749809987
genre_idm: 8.415084591306622
genre_indian: 2.8845074740473855
genre_indie: -8.92244515211582
genre_indie-pop: -10.326995810150189
genre_industrial: -11.5568597510993
genre_iranian: -12.491144984555373
genre_j-dance: -6.294451450443278
genre_j-idol: 6.7174909124398
genre_j-pop: 7.907351347782324
genre_j-rock: 9.047946874371906
genre_jazz: 1.8843720641297814
genre_k-pop: 1.0205022337378686
genre_kids: 20.04372011528289
genre_latin: 2.915485184897995
genre_latino: -3.149902879950684
genre_malay: -8.667526368939043
genre_mandopop: 2.652360948867489
genre_metal: 10.428931696480623
genre_metal-misc: -18.5267059445327
genre_metalcore: 0.18452424134902312
genre_minimal-techno: -17.561395050467773
genre_movies: 3.082082411054585
genre_mpb: -8.454298751441666
genre_new-age: -1.8386578544678525
genre_new-release: 8.275315082549193
genre_opera: 5.136752552292517
genre_pagode: 15.508572335737679

genre_party: 23.930539660584955
genre_philippines-opm: 1.2924358309352622
genre_piano: -9.136081536582724
genre_pop: 29.14624433533526
genre_pop-film: -15.689130764752376
genre_post-dubstep: -15.077985460557814
genre_power-pop: -6.470602322710693
genre_progressive-house: -19.271874789896906
genre_psych-rock: -7.975185715697599
genre_punk: 7.99123724561556
genre_punk-rock: -5.321207978881265
genre_r-n-b: 5.607089608086328
genre_rainy-day: 5.373400074320539
genre_reggae: 10.987782748773785
genre_reggaeton: 16.554447488064948
genre_road-trip: -4.1099321010039755
genre_rock: 30.975026299232418
genre_rock-n-roll: 25.28692719411283
genre_rockabilly: -16.13323515307544
genre_romance: 4.2161709960700495
genre_sad: 15.497008133770503
genre_salsa: 7.414664759857959
genre_samba: 10.08351989606317
genre_sertanejo: 2.426947419705141
genre_show-tunes: 1.82211293680717
genre_singer-songwriter: -9.750742097299774
genre_ska: 2.2760331592955074
genre_sleep: 24.90284272907141
genre_songwriter: -15.98781644921869
genre_soul: 21.506916762223035
genre_soundtracks: -11.494581006838285
genre_spanish: 1.9702694968438266
genre_study: -5.467873234164251
genre_summer: 27.392988101576268
genre_swedish: 9.212741137325786
genre_synth-pop: -7.936450796088995
genre_tango: 9.712948182737252
genre_techno: 7.9671959049274195
genre_trance: 6.817844760558253
genre_trip-hop: -6.608907769710102

genre_turkish: -6.5846207424199195
genre_work-out: -2.1454814018145028
genre_world-music: -5.821062262540982
Plotting feature importances...
Performing residual analysis...
Comparing models...
Training Linear Regression model...
Linear Regression - MSE: 260.71799704388303, MAE: 12.948225129065396, R2:
0.3075907694254485
Training Random Forest model...
Random Forest - MSE: 297.80039569389413, MAE: 13.333507110790443, R2:
0.20910813528342853
Training Gradient Boosting model...
Gradient Boosting - MSE: 286.86283444380035, MAE: 13.796549724176693, R2:
0.23815587443227115
C:\Users\sarbj\AppData\Local\Programs\Python\Python39\lib\site-packages\numpy\ma\core.py:
2846: RuntimeWarning: invalid value encountered in cast
_data = np.array(data, dtype=dtype, copy=copy,
Best Random Forest - MSE: 275.4363016435265
Best parameters: {'max_depth': 30, 'min_samples_split': 10, 'n_estimators': 300}
Scaling numerical features...
Splitting scaled data into train and test sets...
Training Linear Regression model with scaled data...
Mean Squared Error (Scaled Linear Regression): 260.71799704388314
Predicting popularity for example input...
Predicted Popularity: 29.04056253825955
Script executed successfully!